

CD-ROM? Linux uses a file system to manage file storage. It supports various kinds of file systems [2], which are depicted in Figure 1. Fundamentally, a file system is used to organise, manipulate and retrieve stored files from a storage medium. A comprehensive list of file systems, with a brief description of each, can be found at [3] and a comparison of different file systems with distinct features is available at [4].

The Linux file system

Linux used to use the Third Extended file system (ext3) as the default file system, but recent kernel versions (2.6.28 onwards) use the Fourth Extended file system (ext4), which has many more desirable features compared to ext3. The complete features of the ext4 file system are available at [5]. The major differences between ext3 and ext4 are given in Table 1.

File System	Addressing	Maximum file system size	Maximum file size
Ext3	32-bit	16 Terabyte (TB)	2 Terabyte (TB)
Ext4	48-bit	1 Exabyte (EB)	16 Terabyte (TB)

Table 1: Comparison between ext3 and ext4 file systems

Executing `fdisk /dev/<device file name>` will list the specified storage device's partitions, as shown in Figure 2. Type `L` to obtain a list of known partition types, as shown in Figure 3.

Normally, we use the `mount` command to mount a file system, and the `umount` command to unmount the mounted file system. However, we can configure the necessary file systems to be automatically mounted during system start-up, in the `/etc/fstab` file, and thus it is available for use as soon as you log in to the system.

In our day-to-day life, we use different kinds of devices, which in turn use different kinds of file systems. But Linux uses a set of commands to access a file from any device or file system. How is this possible? Linux accesses files through the Virtual File System (VFS), which helps a user to access any device with the same set of commands. VFS completely hides the complexity of different file systems from the user. Figure 4 illustrates how a user accesses a file through VFS. Please refer to Bovet and Cesati [6], to understand the internals of the Linux file system.

The `extundelete` utility [7] can recover deleted files on both ext3 and ext4 partitions. It uses the stored partition's journal information to do this.

Linux uses the `resize2fs` command-line utility to resize (shrink or expand) existing ext2/ext3/ext4 file systems. If you need a graphical utility, then `gparted` [8] can be used.

How is a file accessed by a process? Let us explore this.



Figure 1: Linux-supported file systems

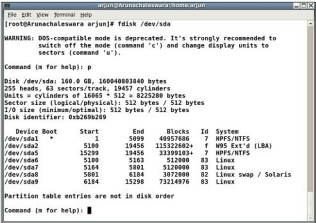


Figure 2: Screenshot of fdisk /dev/sda

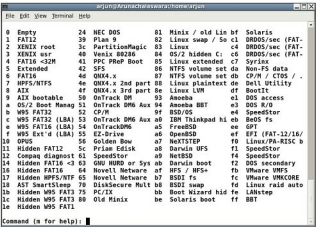


Figure 3: Screenshot-list of partition types